

# EvoSensory

## MINI MANUAL · QUICK START & REFERENCE

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Multi-sensory rhythmic timing training, built by a pediatric OT. This guide covers everything you need to run your first session, understand the data, and document it.

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## THE EVIDENCE

## Why Rhythmic Timing Training Works

Rhythmic timing training has a substantial evidence base built across more than three decades of research in neuroscience, occupational therapy, physical therapy, and special education. The brain's internal timing system — involving the cerebellum, basal ganglia, supplementary motor area, and prefrontal cortex — can be trained through structured, repetitive practice synchronizing movement to an external beat.

When this system is less efficient, the effects are visible in daily life: impulsivity, bilateral coordination difficulty, slow processing speed, reading challenges, handwriting struggles, and poor motor sequencing. EvoSensory measures timing errors to the millisecond and makes them documentable in a way a therapist watching cannot achieve with the naked eye.

| Area                              | Research Finding                                                                                                                                                                                         | EvoSensory Connection                                                                                                                                |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADHD & Attention                  | Children with ADHD show measurable timing deficits independent of inattention. Structured rhythmic training reduces impulsivity and improves sustained attention through cerebellar-prefrontal pathways. | Early-tap tendency + session trend (drifting earlier) map directly onto ADHD timing profiles.                                                        |
| DCD & Bilateral Coordination      | Rhythmic timing training improves alternating limb timing and interlimb synchronization in DCD. Improvements transfer to functional bilateral tasks.                                                     | Late-tap tendency is the characteristic DCD pattern. Levels 1-2 directly train bilateral alternation.                                                |
| Reading & Phonological Processing | Temporal processing underlies phoneme discrimination. Children with dyslexia show significant timing deficits. Rhythmic training improves reading fluency.                                               | Cognitive module — vocabulary spelled by syllable on the beat — directly targets phonological timing.                                                |
| Motor Planning & Praxis           | Sequenced rhythmic training enhances motor program formation and execution speed in praxis deficits.                                                                                                     | Levels 3-5 train increasingly complex sequences. Session trend tracks sequencing under sustained demand.                                             |
| TBI & Acquired Injury             | Post-TBI timing deficits correlate with processing speed and cognitive-motor outcomes. Rhythmic training shows measurable improvement.                                                                   | Session trend drifting late may indicate cognitive fatigue during the session itself — useful diagnostic data.                                       |
| Parkinson's Disease               | Rhythmic Auditory Stimulation (RAS) is one of the most supported non-pharmacological PD interventions — documented improvements in gait cadence and motor timing.                                        | Spatial audio + wired headphones provides the auditory entrainment mechanism central to RAS. Motion module adds gait movement training.              |
| Stroke Rehabilitation             | Rhythmic cuing improves motor timing and bilateral retraining in post-stroke upper and lower extremity rehabilitation.                                                                                   | Motion module (clapping, marching) and unilateral modality settings support targeted stroke rehab protocols.                                         |
| Handwriting                       | Graphomotor timing is one of the strongest predictors of handwriting legibility and speed. Timed rhythmic practice produces transfer effects to handwriting fluency.                                     | Classic levels 1-3 build bilateral and timing foundations that transfer to handwriting. Deviation data directly reflects graphomotor timing quality. |

### **Why millisecond precision matters**

8ms early versus 62ms early is invisible to the human eye but clinically significant. The 8ms patient has an established internal clock. The 62ms patient has a systematic anticipatory pattern that mirrors impulsivity in daily functioning. EvoSensory makes that distinction visible and documentable in every session.

## STEP BY STEP

# Quick Start

**1 Open the app**

Go to **www.evolve-ot.com/evosensory.html** in Chrome. Bookmark it or tap Share → Add to Home Screen for easy access next session.

**2 Create a patient profile**

Tap **+ New Player**. Enter name or initials, select age category (Pediatric, Adult, or Older Adult), and choose training modality. Starting BPM is set automatically. On phones and tablets, most children tap with one hand — select the single-hand option if that applies.

**3 Choose a module and level**

Select **Classic → Level 1**. Adjust BPM with + / −. Use Classic with every new patient.

**4 Confirm hardware**

A setup screen appears before each session. Check what's connected — wired headphones, device vibration, XAC pillow. Only what you confirm is documented in the SOAP note.

**5 Tap with the beat**

Tap the large button, tap the canvas, or press **spacebar** on desktop. One tap per beat.

**6 Add observation and generate SOAP note**

When the session ends, fill in the **Session Observation** field (patient engagement, behavior, fatigue — this populates the S section automatically), then tap **Generate SOAP Note**. Edit before pasting into your EMR.

**🎧 Use wired headphones — most important upgrade**

Spatial audio is the core clinical mechanism — early taps go to the left ear, late taps to the right. Bluetooth adds 100–200ms lag that defeats this. A \$10 wired earbud is all you need.

## Patient Profiles

| Category          | Hands BPM | Feet BPM | Notes                                  |
|-------------------|-----------|----------|----------------------------------------|
| Pediatric (0–17)  | 60 BPM    | 52 BPM   | Standard starting point                |
| Adult (18–64)     | 66 BPM    | 56 BPM   | Slightly faster baseline               |
| Older Adult (65+) | 52 BPM    | 44 BPM   | Wider hit window applied automatically |

Feet start 8–10 BPM slower: hip → knee → ankle → floor is a longer motor sequence than a hand tap. On phones/tablets, verify bilateral hand use or select the appropriate single-hand modality.

## THE FIVE MODULES

## Choosing the Right Module

| Module                                                                                         | What It Trains                                                                                                        | BPM                           | Requires           |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------|
|  Classic      | Core timing, bilateral coordination, spatial audio + haptic feedback, 6 progressive levels. Start every patient here. | Adjustable                    | Any device         |
|  Oculomotor   | Visual tracking synced to beat — pursuits, saccades, depth pulse, gaze hold. Tap-based motor proxy.                   | Adjustable<br>Default: 44 BPM | Any device         |
|  Cognitive    | Vocabulary and math spoken on the beat — phonological processing and number fluency.                                  | Adjustable                    | Any device         |
|  Motion       | Camera detects clapping, knee taps, marching, and jumps timed to the beat.                                            | Adjustable                    | Chrome on computer |
|  Active Games | Body movement controls 5 games — catch stars, pop balloons, kick soccer, duck/jump, balance zones.                    | —                             | Chrome on computer |

**Oculomotor default BPM is 44**

Visual tracking needs more time per beat than hand tapping — the eyes must locate the target, track it, and reset before the next beat. Start at 44 BPM and increase gradually. Adjust with the + / – controls on the module settings screen before starting.

## Classic — Level Progression

| Level | Pattern                | Advance When                                  |
|-------|------------------------|-----------------------------------------------|
| 1     | Alternating L/R        | 85%+ accuracy, 2 sessions                     |
| 2     | Alternating with rests | 85%+ accuracy, 2 sessions                     |
| 3     | Longer held beats      | 80%+ accuracy, stable trend                   |
| 4     | Syncopated patterns    | 80%+ accuracy, <50ms deviation                |
| 5     | Irregular + rests      | 80%+ accuracy, consistent timing              |
| 6     | Full complexity        | Clinical ceiling — maintain for consolidation |

## Motion — Sub-Modes (Chrome on computer required)

Clapping (bilateral UE, waist-up) · Knee Taps/Midline Crossing (waist-head) · Marching (full body) · Jumps (full body + clearance)

## Active Games — Sub-Modes (Chrome on computer required)

Catch Stars (lateral stepping, vestibular) · Pop Balloons (overhead reach) · Kick Soccer (unilateral foot) · Duck & Jump (motor planning) · Side Steps & Balance (static/dynamic balance)

**BPM progression rule**

Increase tempo by 4–8 BPM at a time. Accuracy at a slow, steady tempo is the foundation for everything else. If accuracy falls below 50%, reduce BPM before changing the level.

## READING THE DATA

# Understanding Your Results

## 1. Accuracy — Were they hitting the beat?

% of beats hit within the acceptable time window. Reflects engagement, attention, and task level. High accuracy does not mean good timing.

## 2. Timing Deviation — Where exactly did taps land?

ms (milliseconds) from the beat. Lower deviations are better. Zero = perfectly on the beat.

(+) = early (before the beat) (–) = late (after the beat)

### Accuracy and deviation are not the same thing

A patient can score 95% accuracy but have +55ms average deviation — hitting the window every time, but always early. That is a separate, clinically meaningful finding that accuracy alone does not capture.

## 3. Session Trend — Did timing change during the session?

Tracks whether errors grew, shrank, or held steady as the session progressed. Drifting earlier may indicate difficulty sustaining inhibitory control. Drifting later may indicate attention fatigue.

# Clinical Interpretation Scale

### ACCURACY

| Range   | Interpretation                           | Action                                 |
|---------|------------------------------------------|----------------------------------------|
| 85–100% | Functional range — skill established     | Advance level or increase BPM by 4–8   |
| 70–84%  | Emerging — moderate consistency          | Continue; target 85% across 2 sessions |
| 50–69%  | Below threshold — significant difficulty | Adjust supports; consider slower BPM   |
| <50%    | Well below threshold                     | Reduce BPM 8–10 or drop one level      |

### TIMING DEVIATION (ABSOLUTE VALUE — LOWER IS BETTER)

| Range    | Interpretation            | Clinical Relevance                                   |
|----------|---------------------------|------------------------------------------------------|
| 0–20ms   | Excellent — normal limits | Strong internal clock                                |
| 21–40ms  | Good — mild variability   | Typical in active development                        |
| 41–70ms  | Moderate difficulty       | Likely affecting handwriting, bilateral tasks        |
| 71–100ms | Significant difficulty    | Document functional impact; intervention warranted   |
| 100ms+   | Severe difficulty         | Reduce demands; prioritize sensory feedback supports |

### DIRECTION & SESSION TREND

| Pattern                  | May Indicate                                                                            |
|--------------------------|-----------------------------------------------------------------------------------------|
| Consistently + (early)   | Impulsivity, response inhibition difficulty, anticipatory motor pattern (ADHD, anxiety) |
| Consistently – (late)    | Processing speed delay, motor initiation difficulty, low arousal (DCD, TBI, fatigue)    |
| Stable across session    | Consistent regulation — patient sustained the task without deterioration                |
| Drifting early over time | Mounting impulsivity or difficulty sustaining inhibitory control                        |
| Drifting late over time  | Attention fatigue or processing slowdown under sustained demand                         |

#### Key documentation point

Always report both accuracy and deviation. Connect the deviation direction to a functional impact in the patient's daily occupations — handwriting, turn-taking, dressing, gait, reading rhythm.

## DOCUMENTATION

# SOAP Note Guide

When the session ends, fill in the **Session Observation** field on the results screen — then tap **Generate SOAP Note**. The note is pre-populated with all session data. A "how to read this note" guide is included at the bottom for reviewers unfamiliar with the tool.

## Session Observation field — fills the S section automatically

A text box appears on the results screen before you generate the SOAP note. Type your clinical observation there — patient engagement, behavior, fatigue, any comments made during the session. Whatever you enter populates the S section of the note automatically. If left blank, the S section shows a standard fill-in prompt.

| Section | Content                                                                                                                    |
|---------|----------------------------------------------------------------------------------------------------------------------------|
| S       | Populated from Session Observation field — or blank prompt if not filled in                                                |
| O       | Accuracy %, deviation with direction, session trend, level, BPM, modality, hardware confirmed, mobile caveat if applicable |
| A       | Plain-language interpretation with functional connection prompts and progression status                                    |
| P       | Specific next-session BPM and level recommendation based on this session's data                                            |

## Always edit before use

Draft only. The A section has fill-in prompts — review and personalize before pasting into your EMR or submitting for billing.

# Settings Reference

## Beat Sound

Single Beat · Double Beat · 1-2 Spoken (says "one" and "two" — good for young children)

## Blind Training Mode

Fades visuals after 10s — trains audio/haptic timing only, no visual anchor

## Therapist View

Shows ms deviation and clinical notes in real time during session

## Session Setup Prompt

Hardware confirmation before each session — turn off once workflow is established

## Device Haptics / XAC

Device vibration + XAC pillow bilateral rumble (left = early, right = late)

## Metronome Toggle

Turn off to test internal timing — patient taps without auditory cue

# Hardware & Calibration

| Setup                  | Cost       | Adds                                              |
|------------------------|------------|---------------------------------------------------|
| Any device + browser   | \$0        | Full training, SOAP notes, multi-user profiles    |
| + Wired earbuds        | ~\$10      | Spatial audio: left ear = early, right ear = late |
| + XAC + haptic cushion | ~\$150-200 | Directional whole-body haptic feedback            |
| Chrome on computer     | —          | Motion, Active Games, eye tracking                |

## Eye Tracking Calibration (Chrome on Windows/Mac/Android only)

Enable toggle in Oculomotor settings → tap Play → 9-point calibration grid appears → look at each dot and tap it → tap Start. Gaze crosshair appears on canvas. Accuracy  $\pm 60\text{--}80\text{px}$  (screening-level — not clinical eye movement recording). SOAP note includes appropriate disclosure automatically.

## Camera Positioning (Motion & Active Games)

Positioning guide appears in the setup overlay for each sub-mode. Start button enables automatically when required landmarks are detected in frame. Full-body modes wait until both ankles are visible.